

Coherent Forecasting of Realized Volatility

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Abstract

The QLIKE loss function is the stylized favorite of the literature on volatility forecasting when it comes to out-of-sample evaluation and the state of the art model for realized volatility (RV) forecasting is the HAR model, which minimizes the squared error loss for in-sample estimation of the parameters. An obvious, but rarely implemented progression from the classical HAR model is to align the loss function used for out-of-sample forecast evaluation and in-sample parameter estimation. While the MSE and QLIKE should result in the same solution asymptotically if the model is correctly specified, this empirical work reveals massive forecast performance gains from a HAR model that is directly estimated using the QLIKE loss (henceforth `qlikeHAR`). Especially, if the forecast performance is evaluated using the QLIKE loss, the `qlikeHAR` results in significantly better out-of-sample forecasting performance relative to the classical `mseHAR`.

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1 Introduction

The heterogeneous autoregressive model introduced by [Corsi \(2009\)](#), henceforth HAR) is the state of the art model for forecasting realized volatility (RV) and has seen a myriad of extensions throughout the last two decades (see, for example, [Andersen et al., 2007](#); [Bollerslev et al., 2016](#); [Patton and Sheppard, 2015](#); [Buccheri and Corsi, 2021](#); [Cipollini et al., 2021](#)). All these extensions aim at obtaining statistical evidence to conclude superior forecasting performance over existing approaches for out-of-sample realizations. For this, some loss function is chosen and tests for equal predictive ability in the spirit of [Diebold and Mariano \(2002\)](#) and [West \(1996\)](#), henceforth DMW) are conducted. Since these tests evaluate the forecast errors, their performance depends on the choice of a loss function, which scores the forecast performance. As forecasting volatility is notoriously difficult, empirically one often observes insignificant results of the DMW tests based on the conventional squared error loss function, even in large samples. Naturally, forecasters who investigate the relative performance of different forecasting models strive for a powerful test in order to reveal statistically significant differences and thus opt for the use of a loss function that helps to reject the null hypothesis of equal predictive ability.

For the specific problem of volatility forecasting, [Patton and Sheppard \(2009\)](#) study the efficiency of DMW tests of equal predictive ability depending on the employed loss functions in simulations. As possible candidates for these loss functions, the authors use members of the class of strictly consistent loss functions introduced in [Patton \(2011\)](#) — this class builds upon the so-called class of Bregman losses introduced by [Savage \(1971\)](#) and [Gneiting \(2011\)](#) and nests infinitely many valid loss functions for general mean forecasting but also contains the QLIKE and MSE loss function. The numerical results of [Patton and Sheppard \(2009\)](#) imply that the QLIKE loss is a better choice than the MSE loss, as it results in higher rejection rates under the alternative by maintaining similar size properties. This is supported by many empirical applications; see [Hansen and Lunde \(2006a\)](#) and [Liu et al. \(2015\)](#) among others. As a consequence, the literature on volatility forecasting stylized the QLIKE loss function to its favorite for *out-of-sample* evaluation and many of the newly proposed forecasting models are mainly evaluated in terms of the QLIKE loss; see, for example, [Patton and Sheppard \(2015\)](#); [Conrad and Kleen \(2020\)](#); [Patton and Zhang \(2022\)](#), where QLIKE-based evaluation is performed in the main paper and MSE results are – if at all – presented in the appendix. In addition, using a specific loss function (e.g. QLIKE) is in line with theoretical work of [Patton \(2020\)](#), who claims that

in practical examples with imperfect models the use of a single, predefined, strictly consistent loss function is essential to elicit a forecaster’s best prediction, even if multiple consistent loss functions are available.

Despite the plethora of extensions of the HAR model, the popularity of the QLIKE loss function is not reflected when it comes to the *in-sample* estimation of model parameters, which are – to the best of our knowledge – mostly obtained by minimizing the squared error. In line with the theoretical results in [Patton \(2020\)](#), we expect that using the same loss function for both training the model and evaluating its performance helps to generate better volatility forecasts under a predefined loss function. Hence, we propose to directly estimate the HAR parameters by employing the loss function that is later used for evaluation. That is, one should estimate the parameters based on the QLIKE loss (`qlikeHAR`), instead of the conventionally used squared error loss (`mseHAR`), when the evaluation criterion is QLIKE. Conversely, one should use the conventional `mseHAR`, when the evaluation is based on the squared error loss. To generalize the estimation step, we express the HAR model estimation as an application of a general M-estimator which provides the theoretical foundation to use the QLIKE loss function and any other member of the Bregman class. In general, we refer to a forecasting procedure as *coherent* if the loss function used for in-sample estimation coincides with that used for out-of-sample evaluation.

While the asymptotic solution from both loss functions is generally equivalent for *correctly* specified models ([Dimitriadis et al., 2024](#)), the finite sample estimates and forecasts can differ remarkably. Moreover, the HAR model is primarily used to approximate the long-memory lag structure of realized volatilities which inherently leads to misspecified models. In this case, the parameter estimates from optimizing different loss functions converge to different sets of pseudo-true parameters. Consequently, the parameter estimates optimized for an asymmetric loss function like QLIKE that more heavily penalizes under-predictions should also provide better forecasts if under-predictions in the out-of-sample period are more heavily penalized.

Despite being a straightforward extension to the classical HAR model, the here proposed `qlikeHAR` has not yet been studied in the classical regression literature yet. While there are approaches that recognize the use of different estimation procedures such as weighted least squares, or M-estimation, there has not yet been a study that focuses on the forecast performance of models such as `qlikeHAR`. For example, [Clements and Preve \(2021\)](#) review the use of different flexible estimation procedures, however, they do not specifically use the QLIKE loss function

for estimating the HAR but instead use the Tukey’s bisquare. Nonetheless, the author request to consider estimating model parameters under a loss function that is coherent with the final application of the forecasts. In contrast to the literature on classical linear regression for RV forecasting, there are machine learning based attempts that involve the QLIKE loss function (see, for example [Moreno-Pino and Zohren, 2024](#); [Reisenhofer et al., 2022](#), who report gains in estimation stability and forecasting performance). However, the aforementioned results disagree with the findings of [Rahimikia and Poon \(2020\)](#), underscoring the need for a more fundamental and theoretically grounded investigation within the HAR framework to clarify the underlying mechanisms at play.

This paper contributes to the literature by empirically illustrating the performance gains of coherent volatility forecasting. We provide a simple and fast algorithm that adjusts the in-sample estimation to a specific loss function from the class of Bregman losses. Then, we evaluate the relative performance of `mseHAR` and `qlikeHAR` in simulated and real world forecasting exercises. The theoretical claims are first investigated in a Monte Carlo experiment generating intra-day data from a one factor or two factor stochastic volatility model. We then compute the realized measures and forecast with the `mseHAR` and `qlikeHAR` over different horizons. We find strong simulation evidence that coherent volatility forecasting can improve the forecasting accuracy and the power of DMW tests. The length of the estimation window does not seem to influence the results substantially. Instead, we find evidence that the forecast horizon determines the potential performance gains of coherent forecasting with a particularly strong performance of the `qlikeHAR` for one-day ahead forecasts.

Our empirical application focuses on the cross section of Dow Jones Industrial Average (DJIA) index stocks from 2003 to 2024. For these stocks, we find a significantly better out-of-sample forecasting performance when the `qlikeHAR` is evaluated with QLIKE. When we evaluate with the MSE loss function, the `mseHAR` sometimes results in better forecasts but rarely outperforms the `qlikeHAR` significantly. Again, our results do not seem to be determined by the chosen length of the estimation window which leads us to the conclusion that the difference in forecasting performance is not a small sample phenomenon. Our results reflect the well-known fact that DMW tests with QLIKE loss are generally more powerful than those with MSE loss and we can improve on this feature by optimizing the in-sample parameter estimates with respect to the QLIKE loss.

The paper is organized as follows. Section 2 introduces the setup, reviews the state of

the art theory on forecast evaluation, and connects the insights from forecast evaluation with M-estimation for HAR model based forecasting. Section 3 outlines the equal predictive ability testing framework, discusses the simulation experiments and presents the results of our empirical application. Finally, Section 4 summarizes and concludes.

2 Realized volatility forecasting and evaluation

2.1 Setup

We follow the standard setup, where one considers $p(s)$, $s \geq 0$, as a single asset's logarithmic price. The log prices for trading day t are determined by a conventional diffusion, i.e., the stochastic differential equation is given by

$$dp(s) = \mu(s)ds + \sigma(s)dW(s), \quad s \in [t-1, t], \quad (1)$$

with $\mu(s)$ and $\sigma(s)$ respectively denoting the drift and instantaneous volatility process, and $W(s)$ is a standard Brownian motion.¹ The forecasting target is the *latent* integrated variance, defined for a one day normalized unit time interval as $IV_t = \int_{t-1}^t \sigma^2(s)ds$. We assume the diffusion is observed at equally spaced discrete points in time. Let the Δ -period return be defined by $r_{t,i} = p_{t-1+i\Delta} - p_{t-1+(i-1)\Delta}$, and $M = 1/\Delta$, then for an increasing number of intraday observations, i.e. $M \rightarrow \infty$, a consistent estimator for the *latent* IV_t is the realized variance:

$$RV_t \equiv \sum_{i=1}^M r_{t,i}^2. \quad (2)$$

Throughout the paper, we assume $\mathbb{E}[r_{t,i} \mid \mathcal{F}_t] = 0$. This assumption is conventionally placed onto the returns and renders volatility forecasting problems into a mean forecasting problem for the second moment of the returns. The forecaster's information set is given by the Borel sigma-algebra $\mathcal{F}_t = \sigma(RV_{t-j}, Z_{t-j}; j \geq 0)$, where the possibly large $\mathcal{F}_t$ -measurable vector Z_t contains additionally relevant information, such as other realized measures or external predictors. The optimal h -step ahead forecast for RV_{t+h} , subject to $\mathcal{F}_t$, is defined as

$$\widehat{RV}_{t,t+h} \equiv \arg \min_{RV_t^* \in \mathbb{R}^+} \mathbb{E}[L(RV_t^*, RV_{t+h}) \mid \mathcal{F}_t], \quad (3)$$

¹We exclude the possibility of price jumps for the moment. In the presence of jumps, RV is a biased estimator for the integrated variance. Their effect on RV forecasting is later investigated in the simulation experiments.

where L is considered to be a loss function that elicits $\mathbb{E}[\text{RV}_{t+h} \mid \mathcal{F}_t]$ as the optimal solution for forecast $\widehat{\text{RV}}_{t,t+h}$. Hence, the forecaster should predict the feature based on her best belief about tomorrow's cumulative return distribution, typically assumed to lie in the class $\mathcal{P}$, which contains continuous distributions. We throughout consider the existence of multiple rival forecasts and index them using $j = 1, 2, \dots, J$. The j -th rival forecast might be generated based on different information sets and is defined as $\widehat{\text{RV}}_{t,t+h}^{(j)} = E[\text{RV}_{t+h} \mid \mathcal{F}_t^{(j)}]$.

2.2 Forecast evaluation with loss functions

When interested in building a ranking among rival forecasts, one generally aims for a robust ranking. Robustness of the ranking is assured by using a *strictly consistent* loss function which means that (relative to $\mathcal{P}$),

$$\mathbb{E}[L(\mathbb{E}[\text{RV}_{t+h} \mid \mathcal{F}_t], \text{RV}_{t+h}) \mid \mathcal{F}_t] < \mathbb{E}\left[L(\widehat{\text{RV}}_{t,t+h}^{(j)}, \text{RV}_{t+h}) \mid \mathcal{F}_t\right] \quad \forall \widehat{\text{RV}}_{t,t+h}^{(j)} \neq \mathbb{E}(\text{RV}_{t+h} \mid \mathcal{F}_t). \quad (4)$$

Equation (4) implies that the expected loss is uniquely minimized for the true expected value of the h -step-ahead RV and justifies choosing the forecast with the minimal average loss

$$\bar{L}_{T,j} = \frac{1}{T} \sum_{t=1}^T L(\cdot). \quad (5)$$

[Savage \(1971\)](#) and [Gneiting \(2011\)](#) show that for any *strictly* convex function $\phi(\cdot)$, the class of loss functions that possesses the property in (4) (relative to $\mathcal{P}$) is of the form

$$L(\widehat{\text{RV}}_{t,t+h}^{(j)}, \text{RV}_{t+h}) = \phi(\text{RV}_{t+h}) - \phi(\widehat{\text{RV}}_{t,t+h}^{(j)}) - \phi'(\widehat{\text{RV}}_{t,t+h}^{(j)})(\text{RV}_{t+h} - \widehat{\text{RV}}_{t,t+h}^{(j)}), \quad (6)$$

where $\phi'(\cdot)$ is the derivative of $\phi(\cdot)$.

In addition to strict consistency, a crucial property in both regression-model estimation and forecast evaluation is the (positive) homogeneity of the loss function ([Efron, 1991](#); [Patton, 2011](#)). A loss function is said to be homogeneous of degree $b \in \mathbb{R}$ if $L(c\widehat{\text{RV}}, c\text{RV}) = c^b L(\widehat{\text{RV}}, \text{RV})$ for all $c > 0$ ([Nolde and Ziegel, 2017](#)) and homogeneity implies that conclusions drawn from the loss function are ‘unit consistent’. For forecasts that means that a) changing the scale from U.S. dollars to million U.S. dollars or b) converting the currency from U.S. dollars to Euro or c) re-scaling to percentage returns will not alter the previous conclusions. Motivated by these considerations, and exploiting the fact that RV is non-negative, [Patton \(2011\)](#) proposes the

flexible family of homogeneous loss functions

$$L(\widehat{RV}, RV; b) = \begin{cases} \frac{RV^b - \widehat{RV}^b}{b(b-1)} - \frac{\widehat{RV}^{b-1}}{b-1} (RV - \widehat{RV}) & \text{if } b \in \mathbb{R} \setminus \{0, 1\} \\ \frac{RV}{\widehat{RV}} - \log\left(\frac{RV}{\widehat{RV}}\right) - 1 & \text{if } b = 0 \\ RV \left[\log\left(\frac{RV}{\widehat{RV}}\right) - 1 \right] + \widehat{RV} & \text{if } b = 1 \end{cases}. \quad (7)$$

The parameter b determines the shape of the loss function, ranging from symmetric ($b = 2$) to asymmetric, with heavier penalty either on under-prediction ($b < 2$) or over-prediction ($b > 2$). For $b = 2$ the squared error and for $b = 0$ the QLIKE loss emerges. The different shape of the loss function for different choices of b is illustrated in [Figure A1](#).

2.3 The HAR as an M-estimator

A well-known fact in the finance literature is that the autocorrelation function of the squared returns shows a very strong persistence (long memory) that is difficult to capture in econometric forecasting models ([Andersen and Bollerslev, 1997, 1998](#); [Dacorogna et al., 2001](#)). In striving for the optimal RV forecast, [Corsi \(2009\)](#) proposes the linear model

$$RV_t = \beta_0 + \beta_1 RV_{t-1} + \beta_2 RV_{t-1|t-5} + \beta_3 RV_{t-1|t-22} + u_t, \quad (8)$$

where $RV_{t-j|t-h} = \frac{1}{h+1-j} \sum_{i=j}^h RV_{t-i}$, with $j \leq h$, and the i.i.d. error term u_t . It is natural to solve the linear model using OLS, which implies

$$\widehat{\beta}_T^{\text{OLS}} = \arg \min_{\beta} \frac{1}{T} \sum_{t=1}^T [RV_t - (\beta_0 + \beta_1 RV_{t-1} + \beta_2 RV_{t-1|t-5} + \beta_3 RV_{t-1|t-22})]^2, \quad (9)$$

for the vector of parameter estimates $\widehat{\beta}_T = (\widehat{\beta}_0, \widehat{\beta}_1, \widehat{\beta}_2, \widehat{\beta}_3)^\top$. This solution is essentially a special case of the M-estimator

$$\widehat{\beta}_T = \arg \min_{\beta} \frac{1}{T} \sum_{t=1}^T L(\widehat{RV}_t, RV_t), \quad (10)$$

if L is chosen to be the squared error function ([Huber, 1967](#)). Recently, [Dimitriadis et al. \(2024\)](#) set out to connect the loss functions from M-estimation with loss functions used in forecast evaluation. The authors demonstrate that any M-estimator derived from a loss function of the form presented in Equation (6) results in a model strictly consistent for $\mathbb{E}[RV_t | \mathcal{F}_{t-1}]$. This result builds upon [Newey and McFadden \(1994, p.2123\)](#), where the importance of the

strict consistency condition in (4) is emphasized for consistent M-estimation. Nonetheless, the findings of Elliott et al. (2016) and Dimitriadis et al. (2024) indicate that for any *misspecified* parametric model for a conditional mean, M-estimators with different Bregman loss functions can yield different optimal parameters. This discrepancy is not limited to finite samples as the pseudo-true parameters also differ at the population level.² As outlined above, the HAR model is used as an approximation trying to capture the long memory process of RV in a parsimonious way. It is therefore very likely that the HAR model is misspecified. This motivates the use of general M-estimators to estimate the HAR parameters directly, using the final evaluation criterion to tailor the parameter estimates towards this criterion. This approach ensures that the parameter estimates are aligned with the intended evaluation metric, potentially enhancing the model's performance in empirical applications.

To see why different Bregman losses lead to different pseudo-true parameters under misspecification, consider the following: In the HAR setting, let $X_t = (1, RV_{t-1}, RV_{t-1|t-5}, RV_{t-1|t-22})^\top$ denote the vector of regressors in Equation (8), so that the (in-sample) fitted value is $\widehat{RV}_t = X_t^\top \widehat{\beta}_T$. In this setting, the population first-order condition implied by Equation (6) is

$$\mathbb{E} \left[\phi''(\widehat{RV}_t) (\widehat{RV}_t - RV_t) X_t \right] = \mathbb{E} \left[\phi''(\widehat{RV}_t) (\widehat{RV}_t - \mathbb{E}[RV_t | X_t]) X_t \right] = 0, \quad (11)$$

with $\phi''(\widehat{RV}_t) > 0$ for any Bregman loss in Equation (6). For the special cases of the squared error and QLIKE losses, this yields $\phi''(\widehat{RV}_t) = 2$ and $\phi''(\widehat{RV}_t) = 1/\widehat{RV}_t^2$, respectively. Under correct specification, the pseudo-true parameter β^* satisfies $\widehat{RV}_t = X_t^\top \beta^* = \mathbb{E}[RV_t | X_t]$. That is, the inner term of the first order condition in Equation (11), i.e., $(\widehat{RV}_t - \mathbb{E}[RV_t | X_t])$, is zero at β^* . Hence, the weighting function $\phi''(\widehat{RV}_t)$ becomes irrelevant, and all Bregman losses yield the same pseudo-true parameter. Under misspecification, however, this inner term does not vanish, and different choices of $\phi''(\widehat{RV}_t)$ weight the deviation differently. Consequently, different Bregman losses generally lead to different pseudo-true parameters under misspecification.

Notice that the MSE loss function is the only one that provides a closed form solution to the M-estimation problem stated in Equation (10). Hence, for all other loss functions (e.g. QLIKE), the HAR parameters have to be estimated by numerical optimization.³ While this can be computationally intensive, similar problems also arise for quantile regression problems and the

²White (1982) demonstrates, in the case of maximum likelihood estimation, that the estimators of misspecified models converge to well-defined limits. However, these limits differ under alternative loss functions.

³In the remainder of the paper, we use the default optimizer settings in R (optim), i.e., the Nelder-Mead algorithm for a maximum of 500 iterations and a relative convergence tolerance of 10^{-8} .

computational burden can nowadays be handled easily (see [Baur and Dimpfl, 2018](#), for a quantile HAR-related procedure). Also, cleverly setting the starting values equal to the OLS solution helps to drastically reduce the computation time. Algorithm 1 summarizes the procedure to estimate the parameters of the HAR model based on general M-estimation. This estimation procedure directly generalizes towards common HAR model refinements. In [Figure A2](#) in the Appendix, we track the average computation time for the QLIKE estimation and compare it to the closed form OLS estimation. While the QLIKE estimation as expected takes longer to arrive at the solution, the differences to the OLS estimation are marginal for conventional sample sizes, highlighting that the algorithm is easily applicable in practice.

Algorithm 1 Estimation procedure for HAR model using M-estimation

- 1: **Input:** Historical realized volatility data $RV_t, t = 1, \dots, T$
- 2: **Define:** Aggregated realized volatility $RV_{t-j|t-h} = \frac{1}{h+1-j} \sum_{i=j}^h RV_{t-i}$ for $j \leq h$
- 3: **Initialize:** Starting values for $\beta_0, \beta_1, \beta_2$, and β_3 using OLS estimates $\hat{\beta}_{OLS}$
- 4: **Objective:** Minimize the loss function

$$\hat{\beta}_T = \arg \min_{\beta} \frac{1}{T} \sum_{t=1}^T L(\beta_0 + \beta_1 RV_{t-1} + \beta_2 RV_{t-1|t-5} + \beta_3 RV_{t-1|t-22}, RV_t)$$

- 5: **Set:** Loss function L (e.g., QLIKE or another suitable function from the class of Bregman losses)
 - 6: **Solve:** Optimize the problem numerically if closed-form solution is not available, and use OLS estimates $\hat{\beta}_{OLS}$ as starting values for the numerical optimization
 - 7: **Output:** Estimated parameters $\hat{\beta}_T = (\hat{\beta}_0, \hat{\beta}_1, \hat{\beta}_2, \hat{\beta}_3)^\top$
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3 Forecasting exercises

In this section, we analyze the forecasting performance of the qlikeHAR model. Subsection [3.1](#) outlines the implementation details for model estimation and forecast evaluation. Subsection [3.2](#) presents the setup and results of the forecasting exercise based on simulated data, while Subsection [3.3](#) introduces the empirical results for a real data application. All replication material is available from https://github.com/marius-cp/qlikeHAR_replications.

3.1 Competing forecasting models and evaluation procedures

For each of the data examples in this section, we perform two separate horse-races: First, we compare the mseHAR versus qlikeHAR, where the models are set as in Equation (8). Second, as the proposed procedure is readily extendable towards HAR generalizations such as the semi-

variance HAR of [Patton and Sheppard \(2015\)](#), we compare the mseSHAR versus qlikeSHAR. For completeness, the SHAR is given as

$$RV_t = \beta_0 + \beta_1^+ RS_{t-1}^+ + \beta_1^- RS_{t-1}^- + \beta_2 RV_{t-1|t-5} + \beta_3 RV_{t-1|t-22} + u_t, \quad (12)$$

where $RS_t^+ = \sum_{i=1}^M r_{t,i}^2 I(r_{t,i} > 0)$ and $RS_t^- = \sum_{i=1}^M r_{t,i}^2 I(r_{t,i} < 0)$ are the semivariance estimators. All parameters are estimated as stated in Algorithm 1.

In the following, we consider daily, weekly, and monthly forecast horizons. Whereas the forecasting model for the one-day-ahead forecasting exercise matches the classical approach of [Corsi](#), for the weekly and monthly forecasting horizons we follow the approach established in [Bollerslev et al. \(2016\)](#). That is, the forecast is based on direct projection, in which we replace the daily RVs on the left-hand-side of the different models with the weekly and monthly RVs. Further, we analyse the performance for different options for the length of the estimation windows — particularly, we use 250, 500, 750, 1000, 1500, and 2000 days. Our choice is based on prior literature. For example [Liu et al. \(2015\)](#) use 500 days, [Corsi \(2009\)](#) and [Bollerslev et al. \(2016\)](#) use 1000 days, and [Li and Patton \(2018\)](#) use 1500 days. Hence, these values should cover the most commonly chosen window lengths. Obviously, the largest number of out-of-sample forecasts can be produced when a window length of 250 day is specified, and vice versa, the smallest number for a window length of 2000 days. To foster a fair comparison between all window lengths, we unify the evaluation period as specified in the corresponding subsection. Moreover, we consider 1-minute and 5-minute sampling frequencies to compute the RVs.

Concerning the evaluation techniques, we focus primarily on the QLIKE loss function for evaluation, but also present the MSE loss results. Naturally, for the volatility proxy employed we always stick to the LHS of the respective competing models, i.e. when comparing mseHAR versus qlikeHAR based on 5-min RV, we also use 5-min RV to proxy the volatility realization. Based on these loss functions, we perform for each asset the DMW test for equal predictive ability to compare the qlikeHAR and the mseHAR, which is based on the test statistic

$$DMW_T = \frac{\sqrt{T} \bar{d}_T}{\left[\widehat{\text{avar}} \left(\sqrt{T} \bar{d}_T \right) \right]^{1/2}}, \quad (13)$$

with

$$\bar{d}_T = \frac{1}{T} \sum_{t=1}^T \left[L(\widehat{RV}_{t,t+h}^{(\text{qlikeHAR})}, RV_{t+h}) - L(\widehat{RV}_{t,t+h}^{(\text{mseHAR})}, RV_{t+h}) \right]. \quad (14)$$

Throughout the paper, all test decisions are made at a nominal significance level of 5%. The evaluation is conducted analogously for the SHAR. The asymptotic variance of the average loss difference is computed using the [Newey and West \(1987\)](#) variance estimator with the number of lags set to ten which is inspired by [Patton and Zhang \(2022\)](#). In addition to the DMW tests for the individual assets, we implement a panel version of the DMW test for all constituent stocks of the DJIA. In this case, we estimate the asymptotic variance by adjusting for the individual asset clusters and auto-correlation (again using ten lags). For the implementation, we use the method of [Millo \(2017\)](#), who provides a panel pendant of the Newey-West estimator implemented in the sandwich package of [Zeileis \(2004\)](#).

3.2 Simulated data example

3.2.1 Data generating processes

We start our empirical analysis by studying the potential differences in forecasting performance of the `qlikeHAR` and `mseHAR` in simulation experiments. For this purpose, we simulate intraday price data from one factor and two factor stochastic volatility models according to [Chernov et al. \(2003\)](#), [Barndorff-Nielsen and Shephard \(2004\)](#), [Huang and Tauchen \(2005\)](#), [Bollerslev et al. \(2016\)](#), and others. From these log prices, we compute the daily RV measures, estimate the HAR models and forecast the RV for different horizons. An alternative data generating process is proposed by [Patton and Sheppard \(2009\)](#) who directly generate daily data from a standard GARCH model. However, we aim to use a simulation design that most closely resembles empirical data sets and illustrates the challenges faced in those studies (e.g. the presence of jumps and microstructure noise). If we directly sample daily RV according to a HAR model structure and estimate both `qlikeHAR` and `mseHAR` for a correctly specified model, we find – not surprisingly – that the coefficient estimates converge for large sample sizes. This means that the improvement using coherent estimation and forecasting would be limited to relatively small samples in such a setting. In the following, we focus on RV generated by stochastic volatility models so that their dynamic properties can only be approximated by a HAR model. Hence, in this more realistic case, the HAR model is likely miss-specified with more potential for differences in forecasting performance.

Following [Barndorff-Nielsen and Shephard \(2004\)](#) and [Huang and Tauchen \(2005\)](#), we first draw data from the one factor stochastic volatility model with jumps (SV1FJ). The log prices

are generated according to

$$\begin{aligned} dp(t) &= \mu dt + \exp(\beta_0 + \beta_1 v(t)) dW_p(t) + dJ(t) \\ dv(t) &= \alpha_v v(t) dt + dW_v(t), \end{aligned} \tag{15}$$

where $J(t)$ is a compound Poisson process with constant jump intensity λ and random jump size distributed as $N(0, \sigma_J^2)$. The volatility component is modeled as an Ornstein-Uhlenbeck process with mean reversion parameter α_v . We further allow for constant leverage correlation of the form $\text{corr}(dW_p, dW_v) = \rho$. Due to the presence of a jump component, the price process can have a relatively rugged trajectory. The continuous version of this model, i.e., without a jump component is denoted by SV1F. We also draw from this model and compare the results to those obtained from the SV1FJ to evaluate the effects of jumps on the relative forecasting performance of qlikeHAR and mseHAR.

The two factor stochastic volatility model (SV2F) is also considered in [Chernov et al. \(2003\)](#) and [Huang and Tauchen \(2005\)](#). It is given by

$$\begin{aligned} dp(t) &= \mu dt + \exp(\beta_0 + \beta_1 v_1(t) + \beta_2 v_2(t)) dW_p(t) \\ dv_1(t) &= \alpha_{v1} v_1(t) dt + dW_{v1}(t) \\ dv_2(t) &= \alpha_{v2} v_2(t) dt + [1 + \beta_{v2} v_2(t)] dW_{v2}(t). \end{aligned} \tag{16}$$

Here, a second Gaussian process (v_2) exhibits a feedback term in the diffusion function. In comparison to the SV1FJ, it does not allow for jumps explicitly and the SV2F model has continuous sample paths. Still, it can generate quite rugged price series via the volatility feedback. The leverage correlations are $\text{corr}(dW_p, dW_{v1}) = \rho_1$ and $\text{corr}(dW_p, dW_{v2}) = \rho_2$.

The stochastic differential equations are discretized using a basic Euler-Maruyama scheme. We simulate the log price level for 5000 trading days and then compute the returns at a given sampling frequency, say 1-minute or 5-minute, by taking the difference of the corresponding log price levels. The parameters for both models are chosen in line with [Huang and Tauchen \(2005\)](#) who base their choice on empirical estimates in the literature. We summarize the parameter values in [Table 1](#). The initial values are given by $p_0 = 3$, $v_0 = 0.5$ (SV1FJ) and $p_0 = 3$, $v_{1,0} = 0.5$, $v_{2,0} = 0.5$ (SV2F).

Finally, we investigate the relative forecasting performance in a scenario where market mi-

crostructure noise leads to a departure from the pure semimartingale model. For this purpose, we generate data according to the SV1FJ with our main specification and then add an independently normally distributed error term sequence ϵ with variance $\omega^2 = \{0.001, 0.0001\}$ to the discretized prices. The choice of values is inspired by, for example, [Hansen and Lunde \(2006b\)](#) and [Barndorff-Nielsen et al. \(2008\)](#). The contaminated prices are then given by

$$\tilde{p}_t = p_t + \epsilon_t. \quad (17)$$

Note that the microstructure noise is added to the prices of each intraday interval. This implies that the degree of noise contamination increases for higher sampling frequencies although the variance of ϵ is kept fixed.

Table 1: Parameters for the SV1FJ and SV2F models

	SV1F	SV1FJ	SV2F
μ	0.03	0.03	0.03
β_0	0	0	-1.2
β_1	0.125	0.125	0.04
β_2			1.5
α_v	-0.1	$\{-0.00137, -0.1, -1.386\}$	
α_{v1}			-0.00137
α_{v2}			-1.386
β_{v2}			0.25
ρ	-0.62	-0.62	0.03
ρ_1			-0.3
ρ_2			-0.3
λ		$\{0.058, 0.118\}$	
σ_J		$\{0, 1, 2\}$	

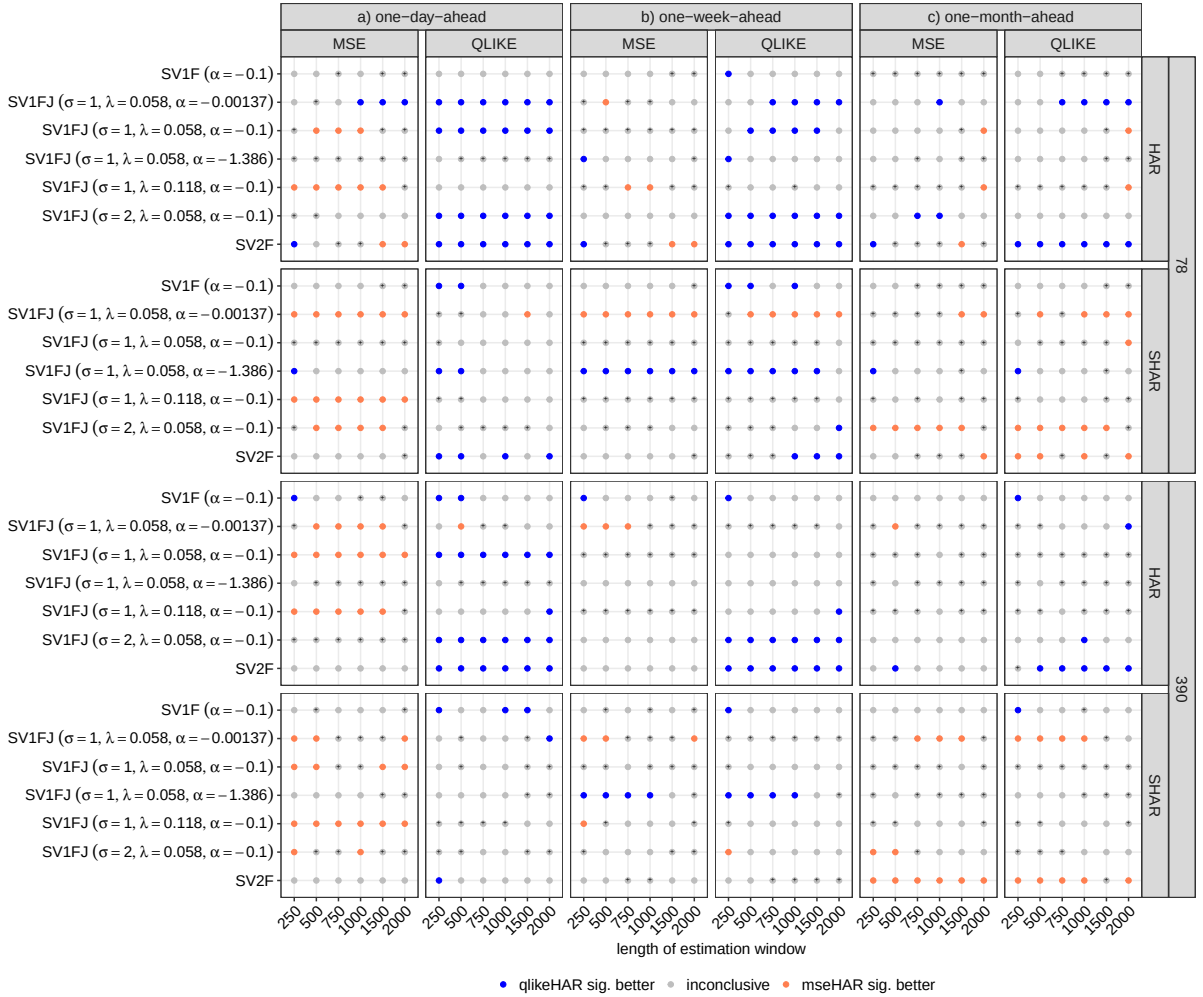
3.2.2 Results

[Figure 1](#) displays the results of DMW tests comparing the forecasting performance of qlikeHAR to mseHAR for either the QLIKE or MSE loss function based evaluations.

For the SV1F, qlikeHAR outperforms mseHAR under QLIKE loss in some cases with mostly inconclusive results under MSE loss. We obtain a clearer picture for 1-minute instead of 5-minute data and estimating the SHAR instead of the HAR model.

The SV1FJ with moderate jump intensity ($\lambda = 0.058$), standard normally distributed jump magnitudes ($\sigma_J = 1$) and moderate mean reversion parameter in the OU process of the volatility component ($\alpha_v = -0.1$) shows the expected pattern of mseHAR outperforming qlikeHAR for one-

Figure 1: Simulation results for different configurations of the stochastic volatility models. The *in-sample* estimation window length is presented on the x-axes. The sampling frequency, 5-minute (78 sampling intervals) and 1-minute (390 sampling intervals), is indicated on the right. Negative test statistic implies better forecasting performance of the **qlikeHAR**. In contrast, a positive test statistic indicates better forecasting performance of the **mseHAR**. The significance level of the DMW test is 5% (blue/coral). A grayish dot indicates an inconclusive DMW test with negative score difference (better **qlikeHAR**). A grayish point which is superimposed by a black + describes an inconclusive but positive DMW test statistic (better **mseHAR**). The upper panel compares the QLIKE and MSE versions of the classical HAR of [Corsi \(2009\)](#). The lower panel considers the QLIKE and MSE versions of the SHAR of [Patton and Sheppard \(2015\)](#).



day ahead forecasts under MSE loss and **qlikeHAR** outperforming **mseHAR** under QLIKE loss for one-day ahead and one-week ahead for 5-minute data. We obtain a similar picture for 1-minute data but **qlikeHAR** only outperforms **mseHAR** significantly for one-day-ahead forecasts under QLIKE loss.

Changing the standard deviation of the jump magnitudes from $\sigma_J = 1$ to $\sigma_J = 2$ (generally larger jumps) positively affects the performance of **qlikeHAR**. Now, **mseHAR** does not outperform **qlikeHAR** anymore under MSE loss if we only consider the standard HAR model. This could be explained by the specific shape of the QLIKE loss function that heavier penalizes under-predictions (see [Figure A1](#)). Large jumps tend to positively affect the RV estimates and the **qlikeHAR** on average predicts larger future RVs than the **mseHAR**. The pattern reverses for an increasing jump intensity parameter moving from $\lambda = 0.058$ to $\lambda = 0.118$ (more jumps). Now, **qlikeHAR** does not outperform **mseHAR** under QLIKE loss but **mseHAR** still outperforms **qlikeHAR** under MSE loss. More jumps do not seem to benefit **qlikeHAR** over **mseHAR** and it appears that **mseHAR** is more robust to an increased jump activity.

If the mean reversion parameter is set to -0.00137 in the SV1FJ, i.e., we simulate data with minimal mean reversion in the volatility function and hence a higher persistence of high volatility states, the **mseHAR** now clearly outperforms **qlikeHAR** under MSE loss at the 1-minute sampling frequency. If we instead set the mean reversion parameter to -1.386 (high mean reversion), we find mostly inconclusive results for both 1-minute and 5-minute data. The **mseHAR** penalizes under- and over-predictions identically and seems to provide more robust forecasts in case of highly persistent volatility.

For the SV2F and 5-minute data, **qlikeHAR** clearly outperforms **mseHAR** under QLIKE loss but the results are more ambiguous under MSE loss. **qlikeHAR** is not as dominant for the SHAR model, particularly in the one-month ahead models. Increasing the sampling frequency from 5-minutes to 1-minute benefits the **qlikeHAR**.

In addition to the out-of-sample results, we also compare the in-sample parameter estimates of both models. Since we directly generate the intra-day data, we cannot evaluate their parameter estimates against a set of “true” HAR coefficients. Therefore, we compare the Euclidean distance from the vector of **mseHAR** to the **qlikeHAR** estimates (see [Figure A3](#)). Our results for the HAR specification suggest that the estimates do not converge for larger estimation windows which implies that both loss functions, although being strictly consistent, lead to a different set of pseudo-true parameters ([Elliott et al., 2016](#); [Dimitriadis et al., 2024](#)). This, in turn, ex-

plains the stable outperformance of the `qlikeHAR` if evaluated under `QLIKE` loss. The average differences for the `SHAR` specification appear to depend more strongly on the length of the estimation window, particularly in the case of the continuous sample paths models `SV1F` and `SV2F`. Otherwise, the `HAR` and `SHAR` results are very similar. [Figure A3](#) clearly shows that the choice of the estimation loss function leads to large differences in coefficients of the one-day ahead models. Conversely, the differences for the one-week ahead model coefficients are much smaller and the one-month ahead coefficients are almost identical. This also translates to the `DMW` test reported in [Figure 1](#) that are mostly insignificant for the one-month ahead models.

The presence of additive microstructure noise contamination does not qualitatively change the main results. While the differences between `qlikeHAR` and `mseHAR` are less pronounced for a higher level of microstructure noise, we still see the same pattern of coherent estimation and forecasting outperforming the corresponding alternative.⁴

In summary, we see no clear pattern of how the window length influences the results. It appears that the benefits of coherent forecasting are not limited to small sample sizes for the in-sample estimation but instead carry over to rather large estimation windows of more than 1,500 daily observations. With such a long sampling period, empirical forecast evaluations also have to take the possibility of structural change into account which naturally limits the overall sample size. Similarly, no clear pattern emerges for the presence of microstructure noise and jump activity. It appears that those aspects increase the noise level in the forecasting exercise but `qlikeHAR` still mostly outperforms `mseHAR` under `QLIKE` loss. In contrast, the forecast horizon seems to be an important factor determining the potential for performance gains of coherent forecasting. Generally, we find that the shorter the forecast horizon, the greater the potential for model misspecification and, consequently, the larger the gains of coherent forecasting approaches.

3.3 Real data example

3.3.1 Data

The `CaPiRe` dataset is a comprehensive and freely available collection of realized quantities related to the Dow Jones Industrial Average Index constituents.⁵ The variables are available at a daily frequency starting from January 2003. The dataset includes the following measures: realized variance, bipower variation, good variance, bad variance, and realized quarticity. All

⁴The results are not included in the paper but can be obtained from the authors upon request.

⁵For further details, visit the `CaPiRe` website at <https://datarepository.stat.unipd.it/item/49>

series are estimated using both 5-minute data (78 observations per day) and 1-minute data (390 observations per day). We compute out-of-sample forecasts for all trading days between 2012-01-01 and 2024-01-01, which corresponds to a total sample size of $T = 3,018$. Notice that we dropped two assets (CRM and DOW) due to missing data during the sampling period. Descriptive statistics reporting the average RV, average bipower variation, and the average jump variation (computed as the difference of both measures) are contained in [Table A1](#) in the Appendix.

3.3.2 Results

[Figure 2](#) presents the results of the real data application using 5-minute RV for parameter estimation and evaluation. We begin with forecast evaluations based on the QLIKE loss function, as represented by the corresponding vertical segments. Using the default predictors of [Corsi \(2009\)](#), located in the horizontal segment labeled HAR, we observe that the `qlikeHAR` outperforms the `mseHAR` in the vast majority of cases under the QLIKE loss function. This conclusion is supported by the predominance of blue-colored points, indicating rejection of the null hypothesis of equal predictive ability at the 5% significance level. Notably, these results remain robust across all forecast horizons considered, although in a handful of cases the strength of the effect diminishes slightly as the horizon increases. To provide further details on the QLIKE evaluation of the HAR models, we consider all assets, forecast horizons, and estimation windows together, resulting in a total of 504 evaluation instances (i.e., $28 \cdot 3 \cdot 6$). Among these, the `qlikeHAR` outperforms the `mseHAR` in 99% of all cases. In addition, the `qlikeHAR` demonstrates significant superiority in forecast performance at the 5% level in 87% of the evaluation cases; see panel a) of [Table 2](#) for a compact summary of these results.

Next, we evaluate the rival HAR models based on the MSE, as illustrated by the respective vertical facets in [Figure 2](#). Here, the results reveal a lower frequency of significant rejections: across the 504 evaluation instances based on the MSE, the `qlikeHAR` significantly outperforms the `mseHAR` in 33% of the cases. Interestingly, these significant losses occur exclusively in one-day-ahead forecasting exercises. The corresponding stocks (“TRV” and “V”) are characterized by a medium average RV and the jump variation is not exceptionally high (see [Table A1](#)).

We now turn to the SHAR results, illustrated in the lower horizontal panel of [Figure 2](#). Among all forecasting horizons, the results for QLIKE based evaluation are consistent with those observed above: if the evaluation is performed using the QLIKE loss function, it is not

Figure 2: This figure reports individual and panel (last row) DMW tests comparing the proposed qlikeHAR versus the baseline mseHAR across 28 Dow Jones stocks based on 5-minute realized volatility. The evaluation is either conducted based on the QLIKE or MSE loss. A negative DMW t-statistic indicates that the qlikeHAR model outperforms the mseHAR model, while a positive t-statistic indicates the opposite. The significance level of the DMW test is 5% (blue/coral). A grayish dot indicates an inconclusive DMW test with negative score difference (better qlikeHAR). A grayish point which is superimposed by a black + describes an inconclusive but positive DMW test statistic (better mseHAR). The upper panel compares the QLIKE and MSE versions of the classical HAR of [Corisi \(2009\)](#). The lower panel considers the QLIKE and MSE versions of the SHAR of [Patton and Sheppard \(2015\)](#).

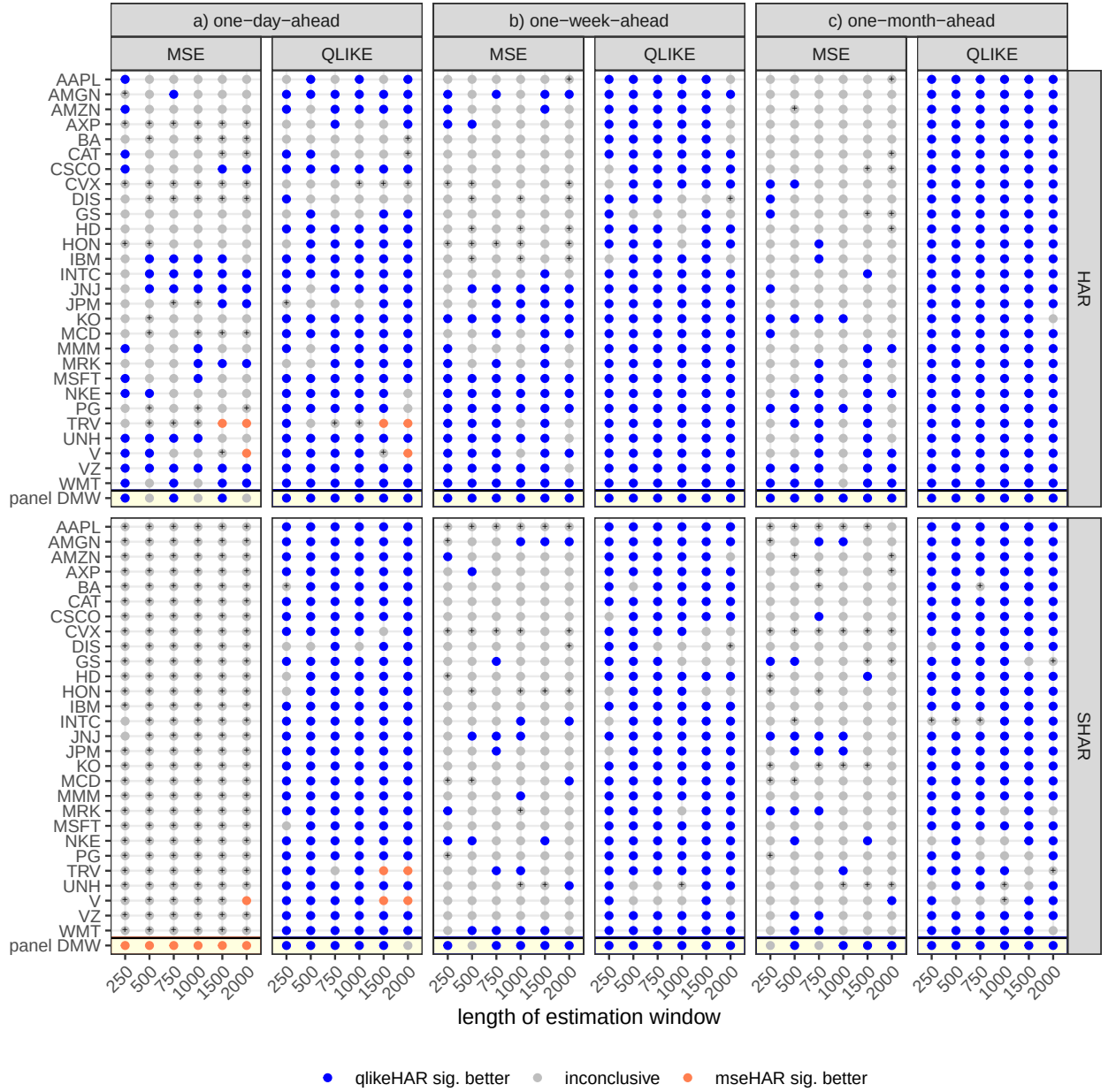
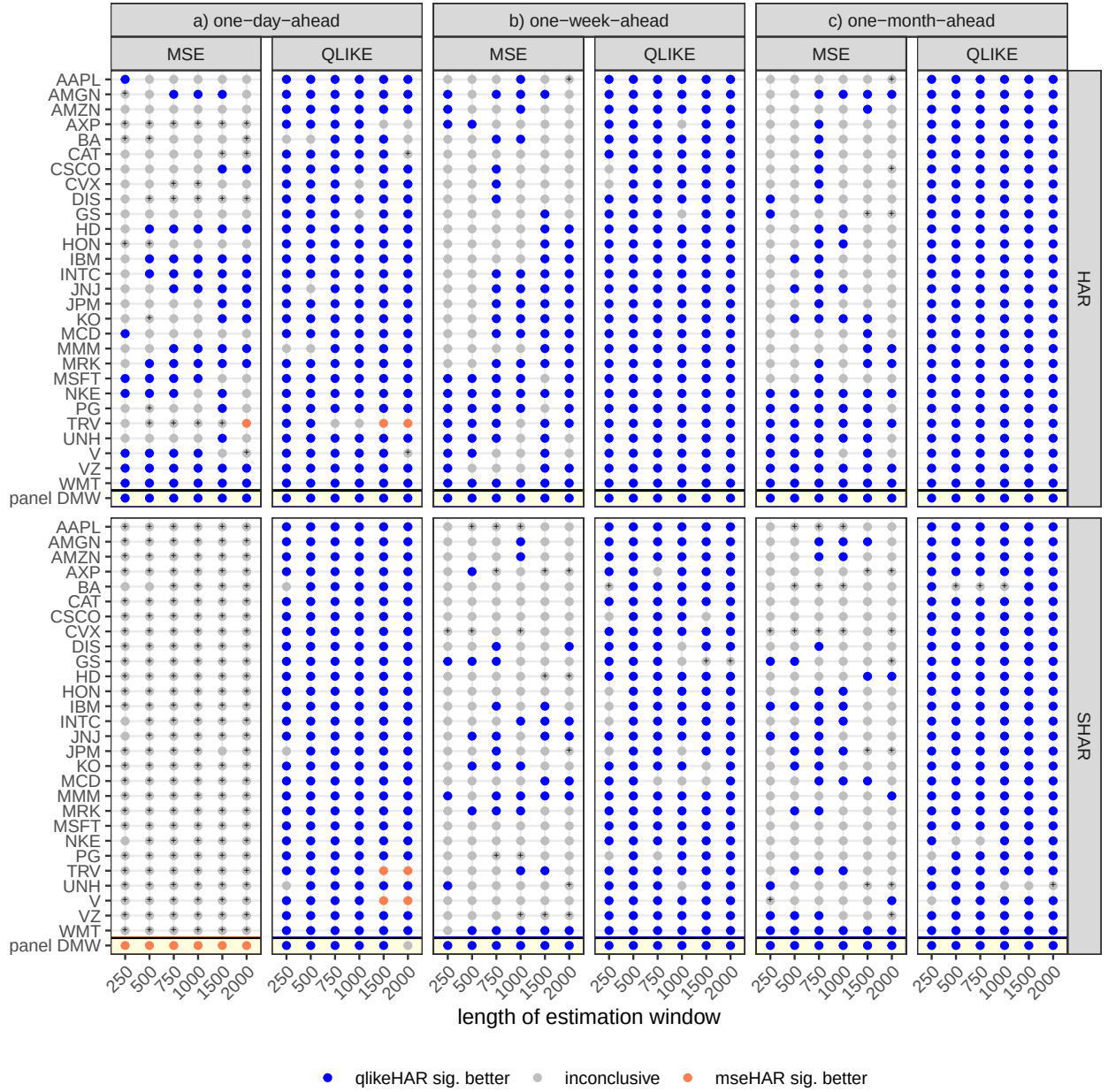


Figure 3: This figure reports individual and panel (last row) DMW tests comparing the proposed qlikeHAR versus the baseline mseHAR across 28 Dow Jones stocks based on 1-minute realized volatility. The evaluation is either conducted based on the QLIKE or MSE loss. A negative DMW t-statistic indicates that the qlikeHAR model outperforms the mseHAR model, while a positive t-statistic indicates the opposite. The significance level of the DMW test is 5% (blue/coral). A grayish dot indicates an inconclusive DMW test with negative score difference (better qlikeHAR). A grayish point which is superimposed by a black + describes an inconclusive but positive DMW test statistic (better mseHAR). The upper panel compares the QLIKE and MSE versions of the classical HAR of Corsi (2009). The lower panel considers the QLIKE and MSE versions of the SHAR of Patton and Sheppard (2015).



recommendable to optimize the MSE for in-sample parameter estimation. In fact, the `qlikeHAR` outperforms the `mseHAR` significantly in 88% (10%) of the evaluation cases when we use the QLIKE (MSE) loss. However, we observe differences when it comes to the MSE evaluation of one-day-ahead forecasts. Here, we clearly see that the `mseHAR` outperforms the `qlikeHAR` for almost every stock but the results are significant at the 5% level only once. Using the panel version of the DMW test, we conclude that the `mseHAR` yield the best forecasting performance for this model.

In [Figure 3](#), we present the results for 1-minute RV data, which are largely consistent with the findings from the 5-minute data. As expected, the DMW tests based on the 1-minute data yield a higher frequency of significant results. For the HAR-based comparisons, the results show that the QLIKE (MSE) metric indicates significantly better performance of `qlikeHAR` in 96% (44%) of the evaluation cases. Similarly, for the SHAR-based comparisons, `qlikeHAR` achieves a higher number of significant wins in 44% and 16% of the cases, according to the QLIKE and MSE metrics, respectively. See also panel b) of [Table 2](#) for a compact summary of these results.

The analysis of real-world data highlights the benefits of consistently using QLIKE loss functions for both in-sample parameter estimation and out-of-sample forecast evaluation. This approach yields superior forecast performance, over all forecast horizons. In contrast, concerning the alignment the use of the MSE loss functions the results appear less clear in terms of forecast performance gains.

		DMW wins		DMW losses	
	RV proxy	Total	Signif	Total	Signif
a) QLIKE based evaluation					
qlikeHAR vs mseHAR	5-min RV	491	436	13	3
qlikeSHAR vs mseSHAR	5-min RV	489	442	15	4
qlikeHAR vs mseHAR	1-min RV	500	484	4	2
qlikeSHAR vs mseSHAR	1-min RV	493	467	11	4
b) MSE based evaluation					
qlikeHAR vs mseHAR	5-min RV	435	169	69	3
qlikeSHAR vs mseSHAR	5-min RV	281	49	223	1
qlikeHAR vs mseHAR	1-min RV	470	222	34	1
qlikeSHAR vs mseSHAR	1-min RV	304	85	200	0

Table 2: Results from 504 Diebold–Mariano–West (DMW) tests comparing HAR models estimated under QLIKE and MSE loss. The evaluation covers 28 Dow Jones stocks using realized volatility computed from 5- and 1-minute intraday returns. Each row lists the number of cases in which the first model (e.g., `qlikeHAR`) attains a lower average loss (“DMW wins”) than its benchmark (e.g., `mseHAR`), with the number of significant cases at the 5% level (“Signif”); “DMW losses” are defined analogously. Results are aggregated across one-day, one-week, and one-month forecast horizons shown in Figures 2 and 3.

4 Conclusion

The results from our simulation experiments and the empirical application clearly show that coherence with respect to estimation and forecasting can improve the RV forecasting performance of HAR models. Particularly, the proposed `qlikeHAR` model which is optimized to minimize the in-sample QLIKE loss, significantly outperforms the competing `mseHAR` in most cases.

The coherent forecasting approach is not limited to the MSE/QLIKE example used in this paper. Generally, it appears useful to first choose a loss function that is later used for forecasting and to design the parameter estimation accordingly (Patton, 2020). In case of the QLIKE, we achieve superior out-of-sample forecasts if we start with an OLS estimation to obtain initial parameter estimates that are then plugged into the M-estimation under the QLIKE loss function. This result holds over a variety of different data-generating processes and different empirical datasets.

The results presented here suggest several promising extensions to related forecasting problems. First, in actuarial prediction contexts, where response variables are typically non-negative and right-skewed, forecasting models—often based on machine learning—commonly employ the Gamma deviance as a loss function (Wüthrich and Merz, 2023). Importantly, the Gamma deviance coincides with the QLIKE loss up to a scale factor. Moreover, as shown in Fissler et al. (2022, Appendix C), classical DMW tests exhibit increased power under such loss structures.

Hence, we expect analogous gains in coherence to those observed for the HAR model in the empirical application of this paper.

Second, the underlying idea of coherent estimation might generalize to other statistical functionals, such as quantiles, where the canonical loss is the (symmetric) piecewise linear-or pinball-loss used in quantile regressions. Forecast evaluation for these functionals can likewise be based on infinitely many strictly consistent loss functions; see [Gneiting \(2011, Theorem 9\)](#). As noted by [Nolde and Ziegel \(2017, Supplement C\)](#), [Gneiting et al. \(2023\)](#), and [Patton \(2020\)](#), applied work in quantile forecasting often favors loss functions with homogeneity (i.e., scale invariance) properties, which can create conflicts between the estimation and evaluation stages. Thus, our coherent estimation principle developed here offers a natural way to reconcile such discrepancies and could thus enhance robustness and interpretability across a broader class of loss functions.

Third, while the present paper concentrates on empirically illustrating the magnitude and practical relevance of the coherence principle in realized volatility forecasting, future research may aim to establish complementary theoretical foundations. In particular, the framework of [Hansen and Dumitrescu \(2022\)](#), who study parameter estimation based on a training sample when the objective is to minimize the expected loss for a distinct evaluation sample, provides a natural starting point for extending the analysis to settings where different strictly consistent loss functions are employed for estimation and evaluation. In parallel, [Mühlemann and Ziegel \(2025\)](#) offer a complementary perspective by showing that the variation in optimal parameters across Bregman losses is itself indicative of model misspecification. Pursuing these theoretical directions could help to deepen the conceptual interpretation of the empirical findings documented here.

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Appendix: Additional material

This appendix provides supplementary figures and tables. In the spirit of Patton (2011), Figure A1 displays the Bregman loss curves for different values of the homogeneity parameter b , introduced in subsection 2.2. Figures A2 and A3 relate to the simulation study presented in subsection 3.2 and report, respectively, the computational time of the qlikeHAR estimator and the average differences between the parameter estimates obtained from the qlikeHAR and the mseHAR. Table A1 shows descriptive statistics for the data used in the empirical study presented in subsection 3.3.

Figure A1: This figure shows the Bregman loss curves for different choices of the homogeneity parameter b . The symmetric MSE loss ($b = 2$) is depicted in green, while the QLIKE loss ($b = 0$), depicted in red, heavier penalizes under-prediction and $b = 4$, depicted in blue, heavier penalizes over-prediction. The true value is set to $RV = 2$ and the forecasts take values from $[0, 4]$. See Figure 1 in Patton (2011) for a similar plot with additional choices of b .

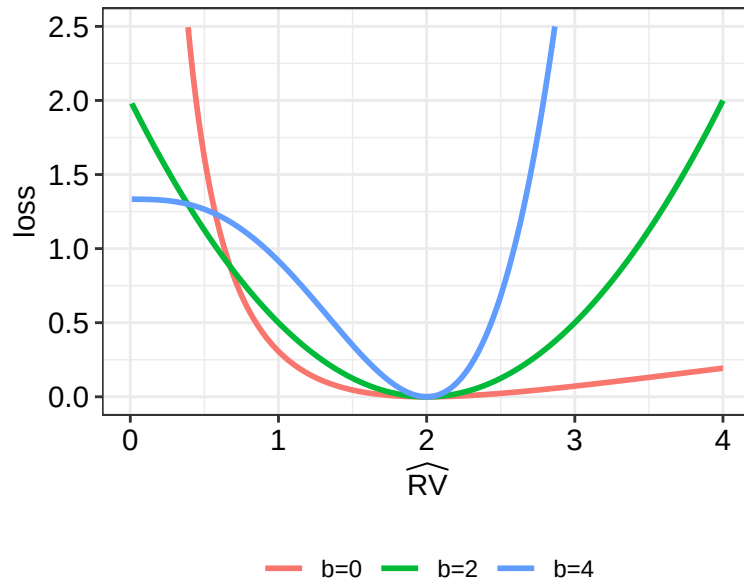


Figure A2: This figure compares the time (in seconds) required for parameter estimation using the classical closed-form OLS procedure and the numerical QLIKE regression method. The y-axis displays time on a logarithmic scale. Outliers are excluded from the plot for clarity.

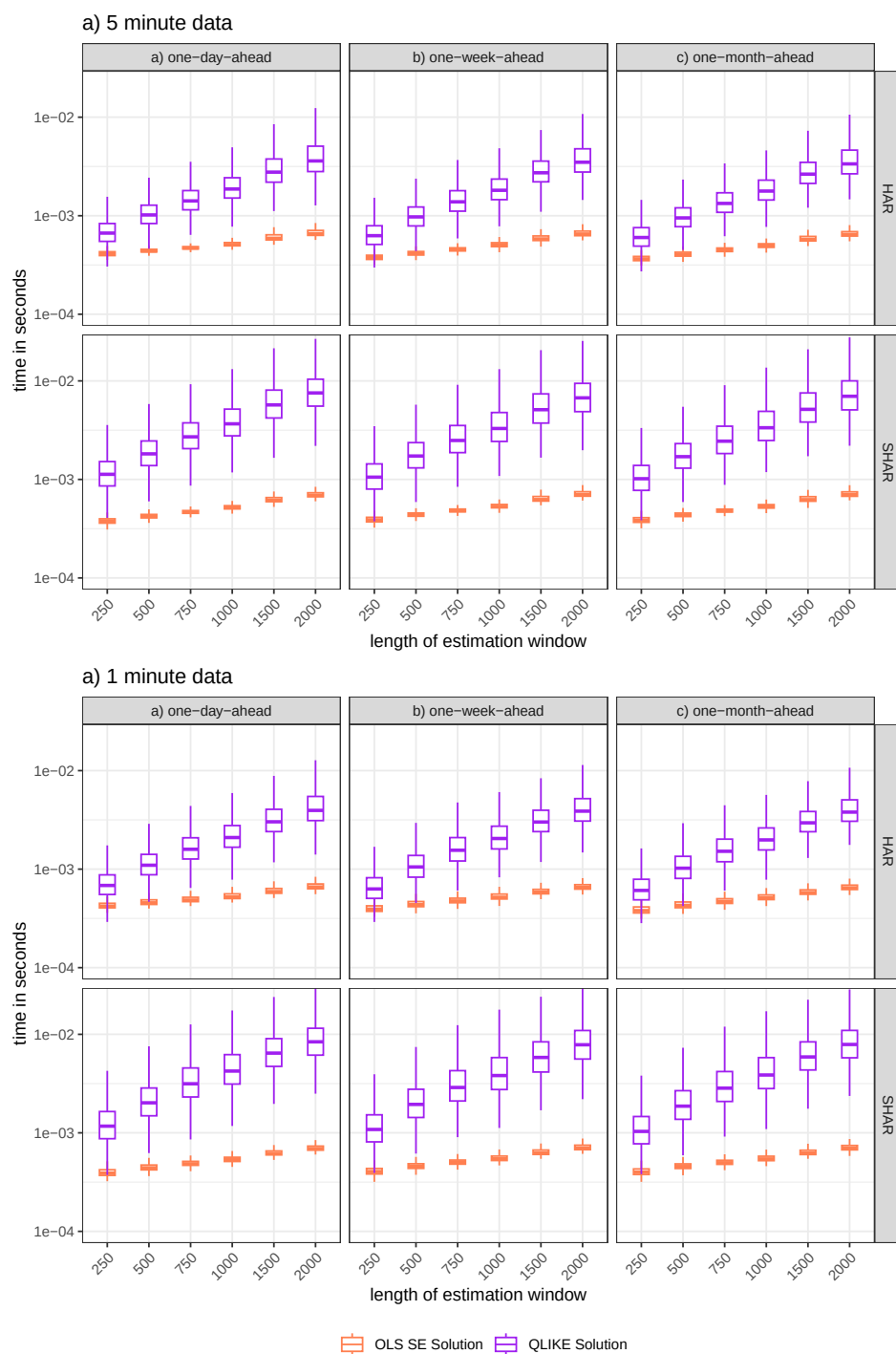


Table A1: This table reports the average RV, average bipower variation (BPV), and average jump variation (JV) for 28 DJIA constituents. The estimates are computed from 5-min and 1-min data between 2011-01-01 and 2024-01-01. Standard deviations are given in parentheses.

	5-minute						1-minute					
	$\overline{RV}$		$\overline{BPV}$		$\overline{JV}$		$\overline{RV}$		$\overline{BPV}$		$\overline{JV}$	
AAPL	2.78	(4.21)	2.61	(4.30)	0.17		2.92	(3.88)	2.77	(3.81)	0.15	
AMGN	2.02	(3.03)	1.83	(2.61)	0.20		2.17	(2.69)	1.98	(2.13)	0.19	
AMZN	3.58	(4.73)	3.36	(4.47)	0.23		3.79	(4.47)	3.61	(4.37)	0.19	
AXP	2.77	(6.22)	2.57	(5.84)	0.20		2.98	(6.62)	2.80	(6.31)	0.18	
BA	2.63	(3.92)	2.45	(3.84)	0.18		2.74	(3.85)	2.59	(3.74)	0.16	
CAT	2.56	(3.34)	2.40	(3.30)	0.15		2.67	(3.51)	2.51	(3.36)	0.16	
CSCO	2.08	(2.48)	1.90	(2.37)	0.18		2.28	(2.58)	2.08	(2.49)	0.20	
CVX	1.88	(2.86)	1.77	(2.86)	0.11		1.98	(2.79)	1.88	(2.76)	0.09	
DIS	1.96	(2.77)	1.81	(2.53)	0.16		2.15	(2.88)	1.99	(2.77)	0.16	
GS	3.09	(10.77)	2.92	(10.22)	0.18		3.30	(11.79)	3.16	(10.97)	0.14	
HD	2.10	(3.49)	1.93	(3.29)	0.16		2.32	(7.00)	2.16	(6.85)	0.16	
HON	1.86	(2.70)	1.72	(2.57)	0.14		1.98	(2.74)	1.81	(2.56)	0.17	
IBM	1.30	(2.11)	1.21	(1.95)	0.10		1.39	(2.17)	1.31	(2.11)	0.08	
INTC	2.46	(2.62)	2.27	(2.46)	0.19		2.66	(2.60)	2.46	(2.48)	0.20	
JNJ	1.00	(3.22)	0.90	(2.71)	0.09		1.06	(2.41)	0.96	(1.98)	0.10	
JPM	3.08	(8.89)	2.88	(8.09)	0.20		3.36	(10.79)	3.19	(10.41)	0.17	
KO	1.03	(1.76)	0.92	(1.43)	0.11		1.12	(1.62)	1.01	(1.50)	0.11	
MCD	1.32	(2.19)	1.19	(1.65)	0.13		1.42	(2.30)	1.27	(1.75)	0.14	
MMM	1.38	(1.93)	1.28	(1.91)	0.10		1.49	(1.93)	1.39	(1.88)	0.10	
MRK	1.82	(4.65)	1.63	(4.00)	0.19		1.88	(3.15)	1.72	(3.15)	0.17	
MSFT	1.79	(2.26)	1.66	(2.14)	0.13		1.94	(2.25)	1.81	(2.22)	0.13	
NKE	2.00	(4.44)	1.84	(3.91)	0.16		2.12	(4.55)	1.90	(3.69)	0.21	
PG	0.99	(1.53)	0.89	(1.41)	0.09		1.04	(1.43)	0.96	(1.40)	0.08	
TRV	1.73	(5.37)	1.55	(4.63)	0.17		1.99	(6.16)	1.77	(5.01)	0.23	
UNH	2.68	(5.80)	2.46	(5.51)	0.22		2.75	(5.48)	2.46	(3.99)	0.28	
V	1.67	(4.07)	1.53	(3.65)	0.14		1.75	(4.08)	1.63	(3.44)	0.12	
VZ	1.53	(3.67)	1.40	(3.22)	0.13		1.73	(5.66)	1.50	(3.68)	0.22	
WMT	1.24	(1.88)	1.12	(1.42)	0.12		1.31	(2.04)	1.18	(1.46)	0.13	

Figure A3: This figure shows the average Euclidean distance between the OLS and M-estimation (QLIKE) coefficient estimates for one-day (red), one-week (green), and one-month (blue) ahead forecasting. The vertical labels refer to 5-minute (78) and 1-minute (390) RVs, respectively.

